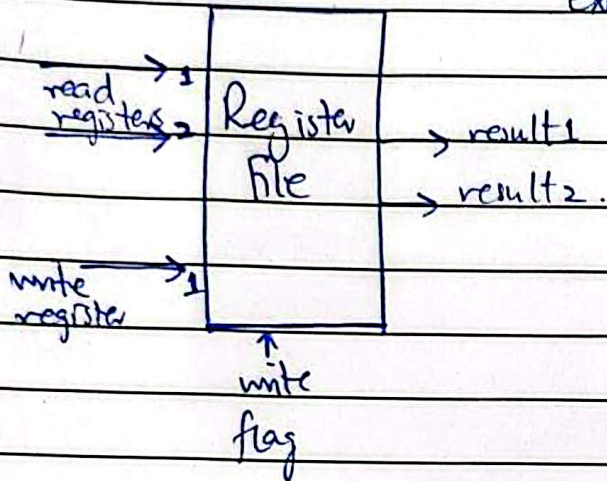


LECTURE-03

Monday
~~Wednesday~~
 24/8/26. #

□ Performance: $\propto \frac{1}{\text{execution time}}$



Inside CPU:

- Datapath
- Control: the sequence of data flow
- cache

⇒ Clock Rate: Cycles per second ~~for frequency~~ (or freq.) $\frac{1}{\text{CCT}}$

- Response Time: Time to complete a particular task
- Throughput: Workdone in a unit time
- Elapsed Time: Total time including all the responses.

(CCT) Clock Cycle/Period: OR $\frac{1}{\text{frequency}}$

$$Q: 250 \text{ ps} = 0.25 \text{ ns} = 250 \times 10^{-12} \text{ s} \rightarrow \text{clock cycle time}$$

$$\text{frequency} = \frac{1}{\text{clock cycle time}} = \frac{1}{0.25} = \boxed{4 \text{ GHz}}$$

NOTE: Hz = cycles / second
 MHz = Million cycles / second
 GHz = Billion cycles / second

- No. of cycles tells how much work CPU needs to do
- Clock Cycle time tells you how long each step takes
- Clock Rate tells how many steps CPU can perform per second

$$\text{Performance} \propto \frac{1}{\text{CPU Execution time}}$$

$$\text{Performance} \propto \frac{\text{Clock Rate}}{\text{No. of Cycles}} \quad \text{Time} = \frac{\text{Clock Cycles}}{\text{Clock Rate}}$$

Q: exec. time of A = 10sec, exec. time of B = 15sec.

$$\frac{\text{Performance of A}}{\text{Performance of B}} = \frac{\text{exec. of B}}{\text{exec. of A}}$$

$$\frac{\text{Perf. A}}{\text{Perf. B}} = \frac{15}{10} = 1.5 \quad \therefore \text{Performance of A is 1.5 times of B (so A is better)}$$

Q: Computer A has 2GHz and time for CPU is 10 seconds.

Computer B has time 6 seconds with 1.2 times as many clock cycles as comp. A. what's the clock rate of B.

$$\text{time} = \text{cycle/rate} \rightarrow \text{Cycles(A)} = 10 \times (2 \times 10^9) = 20 \times 10^9 \text{ cycles}$$

$$\text{Cycles of B} = 1.2 \times (20 \times 10^9) = 24 \times 10^9$$

$$\text{rate of B} = \frac{24 \times 10^9}{6} = 4 \text{ GHz}$$

Date _____

$$\begin{array}{l} \text{CPU Clock} \\ \text{Cycles} \\ (\text{CC}) \end{array} = \begin{array}{l} \text{No. of total} \\ \text{Instruction} \\ (\text{IC}) \end{array} * \begin{array}{l} \text{Average No. of Cycle} \\ \text{per Instruction} \\ (\text{CPI}) \end{array}$$